K-MAPS · DEEP REFERENCE

Qur’ān Vocabulary & Morphology Table Architecture

Every table that stores Qur’ānic vocabulary and morphology across **km_quran** (corpus) and **km_arabic_linguistic** (library) — how data flows from a word to its root, lemma, and context. Grounded in the live D1 state.

مُبِين

Mubeen · manifest

`root ن-ي-ب · Form IV · hollow`

أَنزَلَ

Anzala · sent down

`root ل-ز-ن · Form IV · sound`

كِتاب

Kitāb · the Book

`root ب-ت-ك · noun · sound`

Audited live D1 · **2026-07-10**

Two databases · ~60 tables · three tiers: **root · lemma · context**

0 · The two databases, and the one rule

Two databases, split by a single question — *is this true of Arabic, or true of the Qurʾān?*

	`km_arabic_linguistic` (AL) — the library	`km_quran` (QR) — the corpus
Holds	what is true of **Arabic**: roots, lemmas, morphology, wazn, government	what is true of the **Qurʾān**: occurrences, frequency, render-ready cards
Roots	24,653 total · **1,642 Qurʾānic**	1,642 (agrees with AL)
Lemmas	83,703 total	**4,817 Qurʾānic**
Words	—	77,427 occurrences (99.996% morph-tagged)

> الإفعال (Form IV) is defined by Arabic → **AL**. That أَنزَلَ occurs 183× → **QR**. The corpus never re-states a fact about the language; it stores a **typed reference** (`AL:<id>`) and reads across a Worker service binding.

The storage law (four clauses). Every row obeys these; they are why there are many tables and not one:

1. **Tier (L2).** A fact lives at exactly one scope: **root** (بين) · **lemma** (مُبِين) · **context** (44:2 word 8). A wazn is a *lemma* fact; an أصل واحد is a *root* fact; an iʿrāb sign is a *context* fact.
2. **Once-only (L4).** A root is built once, a lemma once. Passages **reuse**. The 8th time you meet a نزل word, the root and most lemmas already exist.
3. **Provenance (L1).** Every display row resolves to an upstream row via `(source_slug, source_ref)` , **or** declares `is_synthesis = 1` and names its sources. `source_slug LIKE 'kmaps_%'` **means** authored and is never valid with `is_synthesis = 0` .
4. **Core-first (L5).** Display reads a core table; a core table is never empty while a card claims to describe it.

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1 · Simple data-flow — "Word → reuse the root, or add it"



Amortisation. 44:1–8 costs ~30 roots + ~45 lemmas. 44:9–16 reuses most of them. Across the muṣḥaf the marginal cost per verse collapses to the context tier: two lookups and one authored sentence.

2 · Table catalog — every rectangle, with its source

Each table below is a **rectangle**: what it stores · its key · **where its truth comes from** (the provenance rule that governs it). `[AL]` = `km_arabic_linguistic`, `[QR]` = `km_quran`. Row counts are live 2026-07-10.

2.1 · Canonical corpus `[QR]`

```
┌ qr_word_occurrences ──────────────── 77,427 ┐   SOURCE: QAC 0.4 (morphology_tag).
│ one row per word token in the muṣḥaf        │    Natural key: (surah, ayah, word_index).
│ word_text · root · lemma · pos              │    morphology_tag_json holds segments[]/flags/
│ morphology_tag · morphology_tag_json        │    features. This is the ground truth everything reads.
└─────────────────────────────────────────────┘

┌ qr_lemmas ──────────────────────────  4,817 ┐  SOURCE: distinct lemmas of the corpus.
│ lemma_text (UNIQUE) · root · lx_lemma_ref   │  wazn_ar filled 4817/4817 (Gate 3). The join key
│ total_occurrences · wazn_ar                 │  into the form register. lx_lemma_ref → AL lexicon.
└─────────────────────────────────────────────┘

┌ qr_lemma_occurrences ──────────────  72,507 ┐  SOURCE: derived. Maps every lemma to the word
│ lemma_id → (surah, ayah, word_index)        │  tokens that realise it. The reverse of the tag.
└─────────────────────────────────────────────┘
```

2.2 · Root layer `[AL]` core + `[QR]` display

```
┌ ar_ling_roots ──────────── 24,653 (1,642 Qur'ānic) ┐  SOURCE: QAC roots + 13 dictionaries.
│ root_text · root_normalized (UNIQUE, join key)     │   meaning_core_en only 38/1,642 filled —
│ root_type · weak_pattern (1,642/1,642 □)           │  the big remaining authoring gap.
│ frequency_quran · meaning_core_en/ar               │   D31 fixed: NULLs backfilled, dupes merged.
└────────────────────────────────────────────────────┘

   satellites (each keyed by root_id or root_norm):
   • ar_ling_root_semantic_fields   3    field membership (fear, warning…)  ← ar_ling_semantic_fields
   • ar_ling_root_variants          —    orthographic variants              ← lexica
   • ar_ling_root_relations         …    root↔root (antithetical_pair…)     ← a شاهد āyah
   • ar_ling_root_antonyms          12   بشارة↔إنذار                        ← a شاهد āyah (never opinion)
   • ar_ling_root_scholarship       12   Sinai / comparative-Semitic entries  ← ingested chunk id
   • ar_ling_root_sense_axes        6    root × 6 axes × weight(0–3)          ← āyāt scored (basis_json)  [D9 core]
   • ar_ling_root_development_stages 4   ordered semantic evolution           ← Maqāyīs block, when grounded [D9 co
   • ar_ling_root_vocab             56   Membean hook / anchors per root     ← five-lens synthesis

┌ qr_morph_display_roots ──────────────────── 4 ┐  SOURCE: projection of ar_ling_roots + root blocks.
│ id 'MDR:بين' · root_display 'ب ي ن'           │  The render-ready root card. family_json,
│ meaning_core_{ar,en,ur} · family_json         │  senses_json, lens_coverage_json denormalised so
│ senses_json · lens_coverage_json · status     │  the UI needs no join. status: draft→live.
└───────────────────────────────────────────────┘
```

2.3 · Grammatical register [AL] — one dictionary of terms for the whole system

Two tables replace ~14 flat concept tables. Trilingual (ar/en/ur), each term optionally joined to QAC by `qac_flag`.

```
┌ ar_ling_gram_categories ── 24 ┐   pos, tense, voice, transitivity, mood, case,
│ category_key · discipline     │    derivation_type, form_tier, form_family, root_type,
│ (sarf|nahw|balagha|dalala)    │    inflection_class, sentence_class, balagha_* …
└───────────────────────────────┘    SOURCE: authored register (grammar tradition).
┌ ar_ling_gram_terms ──── 245 ┐   'GT:voice:passive' مَجْهُول/passive/مجﻫول, qac_flag='PASS'.
│ id · category_key · term_key  │   parent_term_id builds hierarchy (أضرب الخبر under khabar).
│ qac_flag · name_{ar,en,ur}    │   SOURCE: Shudhūr · Talkhīṣ · QAC-0.4, cited in
│ definition_{ar,en,ur}         │   ar_ling_gram_term_sources.
└───────────────────────────────┘

   • ar_ling_gram_term_relations 14   contrast|subsumes|confused_with (لو/إذا/إنْ)  ← authored
   • ar_ling_gram_term_examples  25   the شاهد for a term                          ← quran/qr_ref
   • ar_ling_gram_term_sources    …   citation per term                            ← page_ref
```

2.4 · Ṣarf engine [AL] — how a lemma is built and governs

```
┌ ar_ling_morphology ──────── 963 bundles ┐  SOURCE: derived from morphology_tag_json.
│ id · pattern · gender/number/case       │  ONE row per distinct QAC parse, SHARED by every
│ tense/voice/person · derivation_type    │  lemma that parses that way. qac_flags is the
│ verb_form · root_type · qac_flags · pos │  natural key; UNIQUE(qac_flags, pos). مُبِين,
│ mood                                    │  مُنذِر, مُرْسِلِين share ACT|MP|ACC|PCPL|(IV).
└─────────────────────────────────────────┘

   • ar_ling_form_paradigms      1,735  (verb_form, root_type) → full paradigm_json  ← wazn table + QAC
   • ar_ling_conjugation_templates 517  person×tense×voice cell forms               ← paradigm, re-keyed (D6)
   • ar_ling_inflection_rules      24   إعلال/إبدال rules (authored once)           ← al-Shāfiya / Ibn ʿUṣfūr
   • ar_ling_verb_government       26   transitivity + حرف (THE irreducible layer)  ← lisān + a شاهد āyah
   • ar_ling_word_sums              1  جذر+وزن+زوائد+i‘lāl decomposition           ← qac_morphology + wazn
   • qr_morph_form_tiers  [QR]      4  ثلاثي/رباعي × تجرّد/زيادة                   ← register (widened 2→4)
   • qr_morph_form_families [QR]   21  الأبواب والصيغ (I–XII + رباعي + noun)       ← register (widened 17→21)
```

`ar_ling_verb_government` is the **only thing QAC cannot give you** and the reason the sarf card can distinguish خَلَصَ, أَخْلَصَ, and أَخْلَصَ لِلهِ. See §3.

2.5 · Lemma layer [AL] core + [QR] display

```
┌ ar_ling_lemmas ──────── 83,703 ┐  SOURCE: lexicon + corpus. is_quran_word flags the 4,817.
│ id · root_id · lemma_text      │  verb_form groups the family. frequency_quran denormalised.
│ part_of_speech · verb_form     │
└────────────────────────────────┘
```

- ar_ling_lemma_root_links 9,280 lemma → root (closes the 4,817) ← qr_lemmas.root
- ar_ling_lemma_morphology 12,431 lemma × attested morphology bundle ← QAC (derived)
- ar_ling_lemma_variants … spelling variants ← lexica
- ar_ling_lemma_registers … register/frequency note ← lexica
- ar_ling_tokens 128,211 one row per QAC segment (prefix/stem/suffix) ← segments[]
- ar_ling_token_lexicon_link … token → lexicon entry ← lx_lemma_ref
- ar_ling_lemma_difficulty 4,817 freq · morph_opacity · polysemy → level(1–5) ← materialised view

```
┌ qr_morph_display_lemmas ──────── 27 ┐  SOURCE: projection of the sarf engine.
│ id 'MDL:{bare}#{lemma_id[:8]}'      │  THE missing tier, now built. Keyed on qr_lemmas.id
│ lemma_ref (UNIQUE) · morphology_id  │  because lemma_bare collides with the root (D30):
│ form_family_id · tier_key · wazn_ar │  بَيِّن & بَيْن both bare to 'بين'. derived_type_*,
│ derived_type_{ar,en,ur} · gloss_*   │  gloss_* trilingual. status draft→live.
└─────────────────────────────────────┘
```

2.6 · Word / context layer [QR]

```
┌ qr_morph_display_words ──────── 24 ┐  SOURCE: Tier-1 strip of the occurrence.
│ (surah, ayah, word_index) UNIQUE   │  Only PROMOTED content words (nouns/verbs/adj) get a
│ surface_ar · lemma_ref → MDL       │  row — particles/pronouns never hit the grid.
│ importance(1–5) · is_anchor        │  No wazn here; that lives at the lemma tier (L2).
└────────────────────────────────────┘

┌ qr_morph_display_blocks ──────── 93 ┐  SOURCE: mixed. THE central display table.
│ scope_level ∈ (root|lemma|context)  │  UNIQUE(scope_level, scope_key, block_type, block_subtype).
│ scope_key · block_type · data_json  │  CHECK enforces provenance: is_synthesis=0 ⇒ real
│ source_slug (FK) · source_ref       │  source_slug + source_ref; is_synthesis=1 ⇒ 'kmaps_%'.
│ is_synthesis · status               │  Root=13 blocks, lemma=3, context=4 (see §5).
└─────────────────────────────────────┘

┌ qr_morph_display_media ──────── 5 ┐  SOURCE: generated/licensed asset in R2.
│ 'MDM:بين:root_dna' · r2_key · sha256 │  Lifecycle pending→stored→live. Linked to a block.
└──────────────────────────────────────┘
```

context blocks read (never copy) from:

- qr_irab_book_entries / iʿrāb display → the irab block
- qr_tafsir_* display → tafsir block (1 per mufassir)
- ar_ling_balagha_examples → balagha block
- qr_translation_sources → translators block

2.7 · Near-synonyms — الفروق [AL]

The kindred/furūq layer. Why ظاهر is not مُبِين; which "send down" verb a verse chose.

```
┌ ar_ling_near_synonym_sets ── 746 (744 draft) ┐  SOURCE: تفريق tradition + LLM draft.
│ set_name · canonical_{en,ar,ur} · review_status│  review_status='draft' blocks citation at
│ 'To descend, to reveal, come down'             │  is_synthesis=0 (D23) — promote before kindred
└────────────────────────────────────────────────┘  can point at them.
   • ar_ling_near_synonym_members    3,269  a lemma in a set + nuance/contrast note  ← member_sources
   • ar_ling_near_synonym_member_sources  …  the claim's citation                    ← source chunk
   • ar_ling_near_synonym_set_sources     …  the set's citation                      ← source chunk
   • ar_ling_near_synonym_evidence        …  a qr_ref شاهد for the contrast          ← qr_word_occurrence
   • ar_ling_near_synonym_review_queue    …  open QA issues                          ← curation
```

2.8 · Verbal idioms & expressions [AL]

```
┌ ar_ling_expressions ──────────── 2,204 ┐  SOURCE: ingested (Islāḥī, tafsīr). NOTE: OCR-noisy,
│ expression_ar · expression_en          │  needs a cleaning pass before display (see gap report).
│ expression_type_id · primary_lemma_id  │  qr_refs_json anchors it to verses.
│ qr_refs_json                           │
└────────────────────────────────────────┘
   • ar_ling_expression_tokens   …  token breakdown of the idiom       ← per-token lemma link
   • ar_ling_expression_types    …  Quranic expression / idiom / …     ← authored, name_{ar,en,ur}
   • ar_ling_collocations        0  lemma-A + lemma-B habitual pairing  ← corpus co-occurrence (EMPTY)
```

The **verb + preposition → meaning** contract (what most people mean by "expression") is stored precisely in `ar_ling_verb_government`, expanded next.

2.9 · Memlet / question learning layer [AL]

```
ar_ling_memlet_kinds 14 · ar_ling_memlets 11 · ar_ling_memlet_media · ar_ling_memlet_mastery
ar_ling_question_kinds 11 · ar_ling_question_items 5 · ar_ling_question_distractors 20
ar_ling_context_passages 3 · ar_ling_exposure_log · ar_ling_vocab_memory_hooks 7
```

Eleven of fourteen memlets are `is_synthesis=0` re-projections of blocks already built (`block_ref` points, never copies). Distractors carry `why_wrong_*` drawn from `form_paradigms` + `near_synonym_members` — every wrong answer is a lesson. Full model in the master doc, Part X.

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3 · Verb government — لازم · متعدٍّ · متعدٍّ بحرف · how many objects

This is the table that turns a root into **different verbs**. QAC tags the form (IV, VIII…) but not *what the verb governs*. `ar_ling_verb_government` stores, per `(root_norm, verb_form)`:

- `transitivity` — the four classes, joined to the register term `transitivity_term_id`
- `harf` — the preposition that completes a `transitive_by_harf` verb ((هـ) / بِ / لـ / من / إلى)
- `meaning_en` — the sense *this government* produces
- `qr_ref` — the شاهد āyah proving it

The four classes (register category `transitivity`):

class	العربية	objects	example (live rows)
`intransitive` (لازم)	لَازِم	0	بَانَ «to become distinct» · نَزَلَ «to come down»
`transitive` (متعدٍّ)	مُتَعَدٍّ	1	بَيَّنَ «to make clear» · كَتَبَ «to write»
`transitive_by_harf`	مُتَعَدٍّ بحرف	1 via حرف	خَلَصَ **إلى** «to reach» · فَسَقَ **عن** «to depart from»
`ditransitive`	مُتَعَدٍّ لمفعولين	2	أَنْذَرَ «to warn [someone] [of something]»

Worked: the نزل family government (real rows). One root, three verbs, three governments:

verb_form	verb	transitivity	حرف	meaning	شاهد
I	نَزَلَ	intransitive	بِ	to come down (bi-: to bring down)	26:193
II	نَزَّلَ	transitive	—	to send down **in stages**	2:23
IV	أَنزَلَ	transitive	—	to send down	44:3 — إِنَّا أَنزَلْنَاهُ

Worked: خلص — why the same root is three verbs. This is the case the whole table exists for:

verb	transitivity	حرف	meaning
خَلَصَ	lazim	—	to become pure
خَلَصَ	lazim	إلى	to reach, arrive at
خَلَصَ	lazim	مِن	to be freed from
أَخْلَصَ	muta'addī	لـ (الدينَ)	to devote (religion) exclusively **to** — 39:2
اِسْتَخْلَصَ	muta'addī	لـ (هـ)	to select exclusively for — 12:54

The حرف *is* the meaning. أَخْلَصَ لِلهِ is not أَخْلَصَ + a note; it is a different verb, and only this table records it.

4 · Per-family expansion — all used words → range of meaning

For any root, the **family** is every attested lemma (masdars, participles, verbs), each with its wazn, and the **range of meaning** the root spans across them. This is `qr_lemmas WHERE root=? ORDER BY total_occurrences DESC`, joined to `wazn_ar`. Two families, live:

بين — «separation → disclosure» (523×, 13 lemmas)

lemma	wazn	occ	derivation / sense in range
بَيْن	فَعْل	266	مصدر — *the between / interval* (the spatial etymon)
مُّبِين	مُفْعِل	119	اسم فاعل IV — **making-clear, manifest** (Mubeen)
بَيِّنَة	فَعِّلَة	71	*clear proof* (the disclosure sense, nominalised)
بَيَّنَ	فَعَّلَ	35	verb II — *to make clear*
تَبَيَّنَ	تَفَعَّلَ	18	verb V — *to become evident* (reflexive)
بَيَان	فَعَال	3	*clear speech / exposition*
… مُسْتَبِين · تِبْيَٰن · تَسْتَبِينَ …	…	1 each	X and other augmentations

Range: the root travels from a *spatial gap* (بَيْن) to *making-visible* (مُبِين, بَيَّنَ) to *proof* (بَيِّنَة). One etymon, a fan of senses — exactly what the `usage_map` and `development` blocks render from `root_sense_axes` + `root_development_stages`.

نزل — «descent → staged revelation» (293×, 13 lemmas)

lemma	wazn	occ	derivation / sense in range
أَنزَلَ	أَفْعَلَ	183	verb IV — **to send down** (Anzala; whole-event)
نَزَّلَ	فَعَّلَ	62	verb II — *to send down in stages* (intensive)
تَنزِيل	تَفْعِيل	15	مصدر II — *the (staged) revelation*
نُزُل	فُعُل	8	*lodging / hospitality* (the settling sense)
تَنَزَّلَتْ	تَفَعَّلَ	7	verb V — *to descend gradually*
نَزَلَ	فَعَلَ	6	verb I — *to come down*
مَنَازِل	مَفَاعِل	2	اسم مكان — *stations / stages (of the moon)*
… مُنزَل · مُنَزِّل · نَزْلَة …	…	1–5	participles + مرّة noun

Range: أَنزَلَ (IV, whole) vs نَزَّلَ (II, in stages) is the theological contrast the family makes visible — and it is *free*, because both are already tagged and their government is one row each in §3.

5 · Three worked words, end to end

5.1 · Mubeen — مُبِين (root بين, Form IV active participle, hollow root)

layer	table	stored value (live)
corpus	`qr_word_occurrences`	root `بين` , lemma `مُبِين` , 119 occurrences
root core	`ar_ling_roots`	root_normalized `بين` , **weak_pattern** `ajwaf_ya` (hollow-yā'), freq 523
root display	`qr_morph_display_roots`	`MDR:بين` , meaning_core_en "*separation and disclosure...*", **status live**
root blocks	`qr_morph_display_blocks` scope=root	**12 blocks**: root_dna, lexicon, derivations, constellation, attestations, kindred, occurrences, usage_map, development, master_story, metaphor, synthesis
lemma	`qr_lemmas`	`مُبِين` , wazn_ar `مُفْعِل`
word sum	`ar_ling_word_sums`	surface `مُبِين` ← underlying `مُبْيِن` , **inflection_class** `ilal_naql` (إعلال بالنقل)
government	`ar_ling_verb_government`	بَيَّنَ (II) transitive «to make clear»; أَبَانَ (IV) transitive
kindred	`ar_ling_near_synonym_sets`	set "*To become manifest, revealed*": مُبِين vs بَدَا · جَهَرَ · ظَهَرَ

The ṣarf lesson: مُبِين and مُنذِر share the *same* wazn مُفْعِل, same Form IV, same اسم فاعل. But ب-ي-ن is **hollow** → مُبْيِن moves to مُبِين by إعلال بالنقل; ن-ذ-ر is **sound** → مُنذِر stays put. *The wazn is constant; the root type decides whether the surface moves.* `word_sums.rule_en` states this contrast verbatim.

5.2 · Anzala — أَنزَلَ (root نزل, Form IV, sound root)

layer	table	stored value (live)
corpus	`qr_word_occurrences`	root `نزل` , lemma `أَنزَلَ` , 183 occurrences
root core	`ar_ling_roots`	root_normalized `نزل` , weak_pattern `sound` , freq 293
root display	`qr_morph_display_roots`	`MDR:نزل` , meaning_core_en "*sending down; revelation sent down in stages*", **status draft**
root blocks	`qr_morph_display_blocks` scope=root	12 blocks (same set as بين)
lemma	`qr_lemmas`	`أَنزَلَ` , wazn_ar `أَفْعَلَ`
government	`ar_ling_verb_government`	**3 rows**: نَزَلَ (I, intr, بِ), نَزَّلَ (II, tr, staged), أَنزَلَ (IV, tr) — the أنزل/نزّل contrast
kindred	`ar_ling_near_synonym_sets`	set "*To descend, to reveal, come down*": نَزَلَ · أَنْزَلَ · تَنَزَّلَ · هَبَطَ · وَضَعَ

D5 resolved. The master doc flagged "*13 ؛ ` ≡≡blocks, no parent root row.*" Live D1 now has `MDR:نزل` with all 12 root blocks — the orphan is fixed; it only awaits promotion `draft → live` .

5.3 · Kitab — كِتاب (root كتب, مصدر/noun فِعال, sound root)

layer	table	stored value (live)
corpus	`qr_word_occurrences`	root `كتب` , lemma `كِتْب` , 260 occurrences
root core	`ar_ling_roots`	root_normalized `كتب` , weak_pattern `sound` , freq 319
root display	`qr_morph_display_roots`	`MDR:كتب` , meaning_core_en "*writing; prescribing; a Book/decree*", **status live**
root blocks	`qr_morph_display_blocks` scope=root	12 blocks
lemma	`qr_lemmas`	`كِتْب` wazn `فِعْل` · family: كَتَبَ (verb I, فَعَلَ), مَكْتُوب, كَاتِب (فَاعِل), اكْتَتَبَ (VIII), (مَفْعُول)
government	`ar_ling_verb_government`	كَتَبَ (I) transitive «to write; to prescribe» — 21:105

Range: كتب spans a *physical act* (كَتَبَ «to write», كَاتِب «scribe», مَكْتُوب «written»), a *legal act* (كُتِبَ «it is prescribed»), and an *object* (كِتاب «the Book / decree»). The noun كِتاب dominates (260/319) — the corpus reaches for the *thing written* far more than the act.

6 · Provenance is the point

Nothing renders that cannot name where it came from. The `qr_morph_display_blocks` CHECK constraint enforces it at write time:

```
CHECK ((is_synthesis = 1 AND source_slug LIKE 'kmaps_%')
    OR (is_synthesis = 0 AND source_slug NOT LIKE 'kmaps_%' AND source_ref IS NOT NULL))
```

A projection (`is_synthesis=0`) must cite a real upstream row (`root_dna` → a Maqāyīs block id). A synthesis (`is_synthesis=1`) must be labelled `kmaps_*` and every strand must resolve. There is no third route. That is why the schema has ~60 tables and not one JSON blob: **a column can be joined, validated, and cited; a blob cannot.**